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A Proofs

In this section, we present our theoretical results on in-context learning Markov Chains with the minimal model of Section 2.3.

Setup and notation Our data consists of sequences of length t, $\boldsymbol{x} = (x_1, \ldots, x_t)$, drawn from a Markov Chain with state space $S = \{1, \ldots, k\}$ (i.e., $x_j \in \{1, \ldots, k\}$ for all $j \in [t]$), and a random transition matrix $\mathcal{P}$. Each row of the matrix is sampled from a Dirichlet distribution with concentration parameter $\boldsymbol{\alpha}$, i.e. $\mathcal{P}_{i,:} \sim \mathrm{Dir}(\boldsymbol{\alpha})$. Unless stated otherwise, we set $\boldsymbol{\alpha} = (1, \ldots, 1)^\top$, corresponding drawing the row from a uniform distribution over the simplex. Let $e_{x_p} \in \{0,1\}^k$ denote the one-hot embedding of the state at position $p \in [t]$ and let $e \in \mathbb{R}^{t \times k}$ be the embedding matrix. We assume there are 2 states in the Markov Chain, i.e. $k = 2$. In that case, the transition matrix can be parameterized as:

$$\mathcal{P} = \begin{pmatrix} a & 1-a \\ 1-b & b \end{pmatrix}, \tag{A.9}$$

where $a, b \sim Unif(0,1)$.

Model We define our model as a simplified sequence to sequence linear transformer $f : \mathbb{R}^{t \times k} \to \mathbb{R}^{t \times k}$ with $f(e) = \mathrm{mask}\left(eW_k(Me)^T\right)e$. It is $W_k \in \mathbb{R}^{k \times k}$ and $M = \begin{pmatrix} v_1 & 0 & \ldots & 0 \\ v_2 & v_1 & \ldots & 0 \\ \vdots & \vdots & \ldots & \vdots \\ v_t & v_{t-1} & \ldots & v_1 \end{pmatrix}$, where $v = [v_1, v_2, \ldots, v_t] \in \mathbb{R}^t$. Equivalently, we can express the i-th logit for the p-th position as:

$$f(e)_{p,i} = \sum_{t'=1}^{p} \mathbb{1}\{x_{t'} = i\} \sum_{s=1}^{t'} v_{t'-s+1} e_{x_p}^\top W_k e_{x_s}. \tag{A.10}$$

The model can represent the unigrams and bigrams solutions as following:

- Construction for bigrams: $v = (0, 1, 0, \ldots, 0)^\top$ and $W_k = I_{k \times k}$, then $f(e)_{p,i} = \sum_{t'=2}^{p} \mathbb{1}\{x_{t'} = i\} \mathbb{1}\{x_{t'-1} = x_p\}$.
- Construction for unigrams: $v = (1, 0, 0, \ldots, 0)^\top$ and $W_k = 11^T$ (all ones), then $f(e)_{p,i} = \sum_{t'=1}^{p} \mathbb{1}\{x_{t'} = i\}$.

Training We analyze stochastic gradient descent training with the margin loss $l_M(f(e)_{p,:}, x_{p+1}) = \frac{1}{k}\sum_{i=1, i \neq x_{p+1}}^{k} \max\left\{0, \Delta + f(e)_{p,i} - f(e)_{p,x_{p+1}}\right\}$. The total loss (sum of the losses across all positions) is given by

$$L(f(e), e) = \left[\frac{1}{t}\sum_{p=1}^{t} l_M(f(e)_{p,:}, x_{p+1})\right], \tag{A.11}$$

and the population loss:

$$\mathcal{L} = \mathop{\mathbb{E}}_{\substack{\boldsymbol{x} \sim \mathcal{P} \\ \mathcal{P} \sim \mathrm{Dir}(\boldsymbol{\alpha})^{\otimes k}}} L(f(e), e), \tag{A.12}$$

where recall e depends on $\boldsymbol{x}$.

We next compute gradients of the loss for the first two steps.

Lemma A.1. *Let the model defined as in eq.* (A.10) *and initialized with* $W_k = c11^T, v = c1^T$. *Then, after one step of gradient descent on the population loss* (A.12) *we have:*

$$\begin{aligned} W_k^{(1)} &= \begin{pmatrix} c & c \\ c & c \end{pmatrix} + c\eta\left[O(t^2)\begin{pmatrix} B & A \\ A & B \end{pmatrix} + O(t)\right] \\ v_j^{(1)} &= c + \frac{c\eta}{t}\left[\frac{(t-j+1)(t-j+2)}{2}D + O(t)\right], j \in [t], \end{aligned} \tag{A.13}$$